

CSC3320 System Level Programming

Homework 1

Due at 11:59 pm on Friday Feb. 23th, 2024

Part I: Answer the following questions

1. Tell the differences between Kernel, API and Shell.
2. Show two examples of the commands using pipe and explain how the pipe works in them separately?
3. In a Linux system, you can issue the command **ls /** to check the sub-directories under root. Please list the names of folders responsible for storing corresponding special files separately.
For example, executable files are mainly stored in “bin”.
 - 1) Shared libraries needed by the programs on the root filesystem.
 - 2) Files used by the bootstrap loader.
 - 3) Device files that help the user interface with the various devices on the system.
 - 4) Mount point for temporarily mounted filesystems.
 - 5) User home directories.
 - 6) Variable data stored by daemons and utilities.
4. Read the article under the link about the differences between Linux and the GNU system <https://www.gnu.org/gnu/linux-and-gnu.en.html> .
And then answer the questions below.
 - 1) What does GNU stand for?
 - 2) What is the GNU project?
 - 3) What are the differences between Linux and GNU system?
 - 4) Is Mac OS X considered to be a BSD Unix or GNU Linux?
 - 5) Is Ubuntu considered to be a BSD Unix or GNU Linux?

Part II: Regular Expression

1. Chose match(es) for each given basic/extended regular expression (maybe multiple correct matches) and describe the pattern of matched string for 3), 4), 6), 11).

e.g. 'ab+a' (extended regex)

a) ababa b) aba c) abba d) aabbaa e) aa

Answer: b,c ; *Pattern : The matched string should begin and end with 'a' and 'b' occurs at least once between leading and ending 'a'*

Note: 1) to 5) are basic regexes;

1) 'a(ab)*a'

(a) ababa (b) aaba (c) aabab (d) aabbaa (e) aa

2) 'a?[bc]*'

(a) abc (b) a?[bc] (c) a[bc] (d) a? (e) abcbc

3) '.[nd]*'

(a) wind (b) window (c) end (d) good (e) will

4) '[a-z]+[0-9]'

(a) a++ (b) aa++ (c) 6+ (d) a+6 (e) aa+1

Note: 5) to 12) are extended regexes.

5) 'a(a|b)*a'

(a) ababa (b) aaba (c) aabab (d) aabbaa (e) aa

6) '[a-z](\+[a-z0-9])+'

(a) x+9 (b) x+y (c) 1 (d) y (e) x++y

7) 'a(bc)*z'

(a) azbc (b) az (c) azbcbcb (d) acc (e) acz

8) 'a.*[0-9]'

- (a) a|1 (b) a*2 (c) ab2 (d) abc (e) aaa

9) '[a-t]+[\\.\\?!+]'

- (a) bash! (b) c (c) Java? (d) python! (e) c++

10) '[a-z]+[\\.\\?!]\\s*[A-Z]' (hint: \\s matches any whitespace characters)

- (a) o. year (b) x?y (c) U.S. (d) Hello!HELLO (e) index?a

11) '(very\\s)?(cool\\s)+(good|bad) weather'

- (a) very weather (b) very cool weather (c) good weather
(d) very cool good weather (e) cool bad weather

12) '-?([0-9])([1-9][0-9])'

- (a) 123 (b) -123 (c) -0 (d) -2.2 (e) 1.2

13) '-?[0-9]*\\.?[0-9]*'

- (a) 123 (b) -123 (c) -0 (d) -2.2 (e) 1.2

2. Write down the extended regular expression for following questions.

E.g. Social security number in the format of 999-99-9999.

Answer: [0-9]{3}-[0-9]{2}-[0-9]{4}

1) 1) Valid date format in the format of **MMM-DD,YYYY**. Note: the first character of the month is capitalized.

(e.g. May-09,2008 Apr-19,2018 Jan-01,2000)

2) An account name which is 8 to 15 characters long, begins with a letter or underscore, ends with a digit, and composed by letters, numbers, underscores and hyphens. (e.g. my_Name1, cat_2, One-1)

3) A single-line comment starting with a # and continuing until the end of the line. Note: only whitespace is allowed between the start of the line and the comment.

(e.g. #This program outputs the string hello world.)

4) Integers in the range between 00 and 99. (e.g. 0, 00, 01, 15, 55, 59,99)

5) Phone number in any of the following format: 999-(999)9999, 999999-9999, 999-999-9999. (Note: all of these formats should be matched by a single regular expression)

Submission:

- Upload an electronic copy (MS word or pdf) of your answer sheet to the folder named “**Homework 1**” of the dropbox in the iCollege system.
- **Please add the homework number and your name at the top of your answer sheet.**
- **Please write down your answers with the question number only in the answer sheet.**
- Name your file in the format of **HW1_FirstnameLastname** (eg. HW1_YuanLong.docx, HW1_YuanLong.pdf)